

Progress Report

Application Of The Wigner Smith Operator To Low Energy Meson Nucleon Interaction

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Abstract

This report investigates the application of the Wigner–Smith (WS) operator to low-energy meson–nucleon interactions within the framework of the Chew–Low model. The analysis is restricted to the one nucleon–one-meson subspace and two interaction models are considered. In the first, nucleon recoil during meson emission or absorption is neglected; in the second, recoil effects are included in the nonrelativistic limit. To build analytical control over both models, the nucleon self-energy correction and the scattering matrix element are computed to second order in perturbation theory.

The second part of the report focuses on the WS operator. After a brief literature review and theoretical overview, the operator is formulated within the restricted Hilbert space. An eigenvalue equation is derived in this setting with the aim of identifying physically meaningful eigenstates and interpreting their connection to the underlying dynamics

1 Introduction

This work represents the first step in my investigation of the applications of the Wigner–Smith (WS) operator. Before embarking on its study in the context of relativistic quantized theories of gravity, I chose to work within a relatively simpler setup. The aim is to develop intuition for the operator’s behavior, its eigen states and eigenvalues, and to extract meaningful physical insights that may increase confidence in the feasibility of applying the WS operator in more complex gravitational settings.

The Chew–Low model was selected as the theoretical framework for two reasons:

1. Familiarity with the model: Through my personal study of relativistic quantum field theory using *An Introduction to Relativistic Quantum Field Theory* by *Silvan S. Schweber*, I encountered the Chew–Low model in detail. A full section of the book is dedicated to it, and I spent time familiarizing myself with the underlying mathematics and physical intuition
2. Although relatively simple, the Chew–Low model provides a good approximation of meson–nucleon interactions in the low-energy regime. Studying the WS operator in this framework could yield results that are either experimentally testable or comparable with more sophisticated theoretical approaches.

2 The Chew Low Model

2.1 Historical Over View

The Chew–Low model was introduced in the mid-1950s as a major development in the understanding of strong nuclear interactions. It proposed an alternative to perturbation theory which, although successful in quantum electrodynamics, is inadequate for the study of strong interactions due to the large value of the coupling constant. By assuming that the mass of the nucleon is much larger than that of the meson, the interaction could be studied in the static source approximation. This essentially means that the nucleon is treated as a fixed source that emits and absorbs mesons without undergoing recoil[1]. Working within this simplified framework enabled the authors to bypass the perturbative approach.

Another key feature of the Chew–Low model is the introduction of a cut-off function. Since the static source approximation relies on a significant mass (and energy) difference between the nucleon and the meson, it is necessary to restrict the meson’s energy. Allowing the meson to gain arbitrarily high energy through

interactions with the nucleon would invalidate the core assumptions of the model. Furthermore, as will be shown later, the energy cut-off is also essential for avoiding divergences in the theory.

The main success of the Chew-Low model is providing an explanation of the P-wave interaction of meson and nucleons [2]. It explained the Δ_{1232} resonance. It also presented cross-symmetry linking different propagation channels together.

2.2 Mathematical Development of the Chew-Low model

The Chew-Low model could be considered as the non-relativistic approximation of the pseudo scalar meson-nucleon interaction, described by the following Hamiltonian.

$$H = \int d^3x : \Psi^*(x)(\alpha.p + \beta M_0)\Psi(x) : + \frac{1}{2} \int d^3x : \Pi.\Pi(x) + \nabla\Phi.\nabla\Phi(x) + \mu_0^2\Phi.\Phi(x) : + \frac{1}{2}G \int d^3x [\Psi(x)^*\tau\gamma_5\Psi(x)].\Phi(x) \quad (1)$$

Here Ψ is the the $SU(2)$ isospin doublet of the proton and neutron, and Φ is the $SU(2)$ triplet of the meson. M_0 and μ_0 are the bare masses of the nucleon and mesons. In this model the $SU(2)$ symmetry between proton and neutron is supposed to be exact. σ (τ) is the Pauli-matrix vector acting on the spin (isospin) of the nucleon.

It can be shown that through a series of canonical transformation[3], this Hamiltonian, could be written as

$$H \approx \int d^3x : \Psi^*(x)(\frac{P^2}{2M_0} + M_0)\beta\Psi(x) : + H_{meson}^{(0)} + \frac{1}{2} \frac{G}{2M_0} \int d^3x \sum_{j=1}^3 [\Psi^*(x)\tau_j\sigma\Psi(x)]\nabla\phi_j(x) \quad (2)$$

In the rest of this work we will limit ourselves to the case where one nucleon is present and is localized at the center of the coordinate system. In the following we introduce a new coupling constant f_0 and we take into account the cut-off function $\rho(x)$, one can then write the Hamiltonian of the system as

$$H = H_0 + H_I \quad (3a)$$

$$H_0 = H_{meson}^{(0)} + H_{nucleon}^{(0)} \quad (3b)$$

$$H_I = \frac{f_0}{\mu_0} \int d^3x \rho(x) \Psi^*(\sigma.\nabla)\tau.\Phi\Psi = \frac{f_0}{\mu_0} \sum_{i,j=1}^3 \int d^3x \Psi^*(x)\sigma_i\tau_j\Psi\partial\phi_j \quad (3c)$$

The free nucleon Hamiltonian takes different form depending on whether the recoil effect is taken into account or not.

1. For the recoil case the Hamiltonian takes the form :

$$H_{nucleon}^{(0)} = M_0 + \frac{P^2}{2M_0} \quad (4)$$

2. In the no-recoil case the Hamiltonian is :

$$H_{nucleon}^{(0)} = M_0 \quad (5)$$

Given that in both recoil and non recoil case the mass of the nucleon appears as a constant in the free Hamiltonian, we change the energy scale so that this constant could be neglected. The cut-off function $\rho(x)$ is taken to be spherically symmetric, i.e., $\rho(x) = \rho(|x|)$. Furthermore the cut-off function is normalized such that

$$\int d^3x \rho(x) = 1 \quad (6)$$

Next we introduce the the Fourier transform of the cut-off function $v(k)^2$

$$v(k^2) = \int d^3x e^{-ik.x} \rho(x) \quad (7)$$

then using the normalization condition of $\rho(x)$, we get $v(0) = 1$.

Given that in all of the following work, only states with 1 nucleon present at all time are considered, the nucleon field operators in the interaction Hamiltonian can be neglected. Thus the interaction Hamiltonian takes the form

$$H_I = \frac{f_0}{\mu_0} \int d^3x \rho(x) (\sigma \cdot \nabla) \Phi(x) \cdot \tau \quad (8)$$

The meson field operators expansion in momentum space is given by :

$$\phi_j(x) = \frac{1}{(2\pi)^{\frac{3}{2}}} \int \frac{d^3k}{\sqrt{2w_k}} (a_{jk} e^{ik \cdot x} + a_{jk}^\dagger e^{-ik \cdot x}) \quad (9)$$

where $w_k = \sqrt{k^2 + \mu_0^2}$ is the energy of the meson and a_{jk} ($a_{jk}^\dagger$) is the annihilation (creation) operator for one meson state with isospin j and 3-momentum k . They respect the following commutation rules

$$[a_{jk}, a_{j'k'}] = [a_{jk}^\dagger, a_{j'k'}^\dagger] = 0 \quad (10a)$$

$$[a_{jk}, a_{j'k'}^\dagger] = \delta_{jj'} \delta^{(3)}(k - k') \quad (10b)$$

Using this expression of the meson operator one can simplify the expression of the interaction Hamiltonian (see appendix for proof):

$$H_I = \sum_{j=1}^3 \int d^3k (V_{jk} a_{jk} + V_{jk}^\dagger a_{jk}^\dagger) \quad (11)$$

Where V_{jk} is given by

$$V_{jk} = \frac{if_0}{\mu_0} \frac{(\sigma \cdot k) v(k^2)}{\sqrt{((2\pi)^3 2w_k)}} \tau_j \quad (12)$$

Next we propose to express the free Hamiltonian of the nucleon in the recoil case, in the center of mass system to this end let

$$P_M = \sum_{j=1}^3 \int d^3k k a_{jk}^\dagger a_{jk} \quad (13)$$

be the total momentum operator for meson states. In the center of mass frame, the total momentum is zero, thus :

$$p = -P_M = - \sum_{j=1}^3 \int d^3k k a_{jk}^\dagger a_{jk} \quad (14)$$

In this case the free Hamiltonian :

$$H_0 = H_{meson}^{(0)} + H_{nucleon}^{(0)} \quad (15)$$

is given by

$$H_0 = \sum_{j=1}^3 \int d^3k \left(\frac{k^2}{2M_0} + w_k \right) a_{jk}^\dagger a_{jk} + \sum_{j,l=1}^3 \int d^3k \int d^3k' \frac{k \cdot k'}{2M_0} a_{jk}^\dagger a_{lk'}^\dagger a_{jk} a_{lk'} \quad (16)$$

2.3 Calculation of the self-energy interaction of the nucleon state

Consider the case of one nucleon system, in absence of the interaction the center of mass frame coincides with the rest frame of the nucleon thus the energy of the system is 0. However due to the interaction with the meson field, the nucleon requires an interaction energy ΔE . The aim of this paragraph is to determine this energy to second order in the coupling constant using perturbation theory.

The application of perturbation theory is based on the fact that the unperturbed and perturbed Hamiltonian have the same spectrum. To satisfy this condition let :

$$H'_0 = H_0 + \Delta E \quad (17a)$$

$$H'_I = H_I - \Delta E \quad (17b)$$

In absence of interaction a one nucleon state is characterized by it's spin, s , and it's isospin, t . It verifies the following eigen state equation :

$$H'_0 |s, t\rangle = \Delta E |s, t\rangle \quad (18)$$

Next considered the physical (dressed) one nucleon state, $|s, t\rangle_s$. In the center of mass system, this state verifies the equation :

$$H' |s, t\rangle_+ = \Delta E |s, t\rangle_+ \quad (19)$$

To take the development further we write the dressed nucleon state in the following form :

$$|s, t\rangle_+ = \sqrt{Z} |s, t\rangle + |X\rangle \quad (20)$$

Where the state $|X\rangle$, verifies $P_n |X\rangle = 0$, where P_n is the projector on 1 nucleon states.

$$P_n = \sum_{s,t} |s, t\rangle \langle s, t| \quad (21)$$

It's than easy to prove (see appendix) that equations 7 lead to :

$$|s, t\rangle_+ = \sqrt{Z} |s, t\rangle + (1 - P_n) \frac{1}{\Delta E - H'_0} H'_I |s, t\rangle_+ \quad (22)$$

Using equations 17, we get :

$$\langle s, t| H'_0 - H' |s, t\rangle_+ = 0 \quad (23a)$$

$$\langle s, t| H'_I |s, t\rangle_+ = 0 \quad (23b)$$

$$\langle s, t| H_I - \Delta E |s, t\rangle_+ = 0 \quad (23c)$$

$$\Delta E \langle s, t|s, t\rangle_+ = \langle s, t| H_I |s, t\rangle_+ \quad (23d)$$

$$\sqrt{Z} \Delta E = \langle s, t| H_I |s, t\rangle_+ \quad (23e)$$

Using the expression for $|s, t\rangle_+$ given by equation 22, and restricting ourselves to second order terms we get

$$\Delta E^{(2)} = \langle s, t| H_I |s, t\rangle + \langle s, t| H_I (1 - P_n) \frac{1}{\Delta E - H'_0} H_I |s, t\rangle \quad (24)$$

Since The effect of H_I is either to create or annihilate a meson state, the inner product $\langle s, t| H_I |s, t\rangle$ is null. Using the same the reasoning we get $P_n H_I |s, t\rangle = 0$. Thus equation 24 simplifies to :

$$\Delta E^{(2)} = \langle s, t| H_I \frac{1}{\Delta E - H'_0} H_I |s, t\rangle \quad (25)$$

Using equation 11, we finally get :

$$\Delta E^{(2)} = \sum_{j,l=1}^3 \int d^3k \int d^3q \langle s, t| V_{jk} a_{jk} \frac{1}{\Delta E - H'_0} V_{lq}^\dagger a_{lq}^\dagger |s, t\rangle \quad (26)$$

Using the fact that (proof in the appendix):

$$a_{jk} \frac{1}{\Delta E - H'_0} = \frac{1}{\Delta E - H'_0 - \hat{f}(k)} a_{jk} \quad (27)$$

With

$$\hat{f}(k) = w_k \text{ For the now recoil case} \quad (28a)$$

$$\hat{f}(k) = w_k + \frac{k^2}{2M_0} + \frac{k \cdot P_M}{M_0} \text{ For the recoil case} \quad (28b)$$

With P_M is the meson total momentum operator given by equation 14. Injecting this to equation 26, we get

$$\Delta E^{(2)} = \sum_{j,l=1}^3 \int d^3k \int d^3q \langle s, t| \frac{1}{\Delta E - H'_0 - \hat{f}(k)} V_{jk} a_{jk} V_{lq}^\dagger a_{lq}^\dagger |s, t\rangle \quad (29)$$

Consider next

$$\frac{1}{\Delta E - H'_0 - \hat{f}(k)} |s, t\rangle = -\frac{1}{f(k, 0)} |s, t\rangle \quad (30)$$

with

$$f(k, k') = w_k \text{ (No - recoil)} \quad (31a)$$

$$f(k, k') = w_k + \frac{k^2}{2M_0} + \frac{k \cdot k'}{M_0} \text{ (Recoil)} \quad (31b)$$

Using this definition in equation 29, we get

$$\Delta E^{(2)} = \sum_{j,l=1}^3 \int d^3k \int d^3q - \frac{1}{f(k, 0)} \langle s, t | V_{jk} a_{jk} V_{lq}^\dagger a_{lq}^\dagger | s, t \rangle \quad (32)$$

Using the commutation rules 10, we get :

$$\Delta E^{(2)} = \sum_{j=1}^3 \int d^3k - \frac{1}{f(k, 0)} \langle s, t | V_{jk} V_{jk}^\dagger | s, t \rangle \quad (33)$$

Using the proprieties :

$$(\sigma \cdot k)^2 = k^2 \quad (34a)$$

$$\tau_j^2 = 1 \quad \forall j \in [1, 2, 3] \quad (34b)$$

we get :

$$V_{jk} V_{jk}^\dagger = \frac{f^2}{\mu_0^2} \frac{k^2 v(k^2)^2}{2w_k (2\pi)^3} \quad (35)$$

Thus :

$$\Delta E^{(2)} = -\frac{3f^2}{2\mu_0(2\pi)^3} \int d^3k \frac{k^2 v(k^2)^2}{w_k f(k, 0)} \quad (36)$$

Given that for both recoil and no-recoil cases $f(k, 0)$ depend only on the norm of the momentum k , we can conduct the angular calculation to get:

$$\Delta E^{(2)} = -\frac{3f^2}{4\mu_0\pi^2} \int_0^{+\infty} dk \frac{k^4 v(k^2)^2}{w_k f(k, 0)} \quad (37)$$

Using the value of $f(k, 0)$ for both recoil and no recoil case we get :

$$\Delta E^{(2)} = -\frac{3f^2}{4\mu_0\pi^2} \int_0^{+\infty} dk \frac{k^4 v(k^2)^2}{w_k^2} \text{ No-recoil} \quad (38a)$$

$$\Delta E^{(2)} = -\frac{3f^2}{4\mu_0\pi^2} \int_0^{+\infty} dk \frac{k^4 v(k^2)^2}{w_k (\frac{k^2}{2M_0} + w_k)} \text{ Recoil} \quad (38b)$$

Consider next the point like interaction case for which $v(k^2) = 1$, then taking the high momentum limit for which $w_k \approx k$, that the the no recoil case is quadratically divergent while the recoil case make the the self energy linearly divergent. Already at this stage the problem of infinite self energy arise, however the model was build in the non relativistic case which is only valid for low energies hence the importance of having a cut-off function to preserve the consistency of the non-relativistic approximation.

2.4 Calculating the Scattering Amplitude for the chew low model

In this section, we aim to calculate, to second order in perturbation theory, the scattering amplitude for the process $N + \pi \rightarrow N + \pi$. The calculation is carried out, as in the previous paragraph, in parallel between recoil and no recoil cases. And the differences is emphasizes at the end. It's worth saying that, following the open remarks made at the start of the Chew-Low model discussion, perturbation theory is not effective in treating this process. Indeed the results found are not into big accordance with experiment, yet this is still

high valuable exercise, given that most of the techniques used in this paragraph are helpful to our study of the Wigner Smith operator in the the rest of this work.

Consider the initial and final states $|\Phi_i\rangle$ and $|\Phi_f\rangle$ where :

$$|\Phi_i\rangle = |j, k, s, t\rangle = a_{jk}^\dagger |s, t\rangle \quad (39a)$$

$$|\Phi_i\rangle = |l, q, s', t'\rangle = a_{lq}^\dagger |s', t'\rangle \quad (39b)$$

$$(39c)$$

To second order the transition matrix is given by :

$$R_{fi} = -\langle l, q, s', t' | \Delta E^{(2)} | j, k, s, t \rangle + \langle l, q, s', t' | H_I \frac{1}{E_i - H_0 + i\epsilon} H_I | j, k, s, t \rangle \quad (40)$$

Note that the initial and final subspaces are the same, i.e, one meson, one nucleon subspace. So one should only consider the part of the operator $H_I \frac{1}{E_i - H_0 + i\epsilon} H_I$ that leaves this subspace invariant. Put differently if the first interaction Hamiltonian creates a meson states the second should annihilate a meson state and vise vers ca. The second term of equation 40, is therefore could be written as :

$$\begin{aligned} \langle l, q, s', t' | H_I \frac{1}{E_i - H_0 + i\epsilon} H_I | j, k, s, t \rangle &= \sum_{m,n=1}^3 \int d^3 p_1 \int d^3 p_2 \langle \Phi_f | a_{mp_1} V_{mp_1} \frac{1}{E_i - H_0 + i\epsilon} a_{np_2}^\dagger V_{np_2}^\dagger | \Phi_i \rangle + \\ &\quad \langle \Phi_f | a_{mp_1}^\dagger V_{mp_1}^\dagger \frac{1}{E_i - H_0 + i\epsilon} a_{np_2} V_{np_2} | \Phi_i \rangle \end{aligned} \quad (41)$$

This term could be written as (proof in appendix) :

$$\langle l, q, s', t' | H_I \frac{1}{E_i - H_0 + i\epsilon} H_I | j, k, s, t \rangle = -L(q) \delta_{jl} \delta_{ss'} \delta_{tt'} \delta^{(3)}(k - q) + \langle s', t' | \frac{V_{lq}^\dagger V_{jk}}{f(q, q)} - \frac{V_{jk} V_{lq}^\dagger}{f(k, q)} | s, t \rangle \quad (42)$$

with :

$$L(q) = \frac{3f^2}{2(2\pi)^3 \mu_0^2} \int d^3 p \frac{v(p^2)^2 p^2}{w_p f(p, q)} \quad (43)$$

Thus the transition matrix on the energy shell is given by :

$$R_{fi} = -(\Delta E^{(2)} + L(q)) \delta_{jl} \delta_{ss'} \delta_{tt'} \delta^{(3)}(k - q) + \langle s', t' | \frac{V_{lq}^\dagger V_{jk}}{f(q, q)} - \frac{V_{jk} V_{lq}^\dagger}{f(k, q)} | s, t \rangle \quad (44)$$

The 4 terms appearing in this equation could be shown in manner similar to Feynman diagrams :



Diagram (a) corresponds where the term ΔE acts alone. Diagram (b) corresponds to the process where the initial meson is first absorbed and the final meson is emitted, while diagram (c) corresponds to the opposite process where the final meson is first emitted and then the initial meson is absorbed. Diagram d corresponds to the case where the nucleon emits and reabsorbed the meson.

for The no recoil case one $f(p, q) = w_p$ so that $L(q) = -\Delta E^{(2)}$ and thus the first term in equation 44 conceals out. This is due to the fact that the self energy of the nucleon to second order is due to the emition and reabsorption of a meson. In sense the self energy term of the 1 meson 1 nucleon state is the same as the one nucleon state. thus is because recoil is neglected so that the energy of nucleon is unchanged while emitting or absorbing a meson. In the recoil case on the other hand, the self energy of the one meson one

nucleon state is different from that of the nucleon only state. Expanding the expression of the operator V we get for the no recoil case

$$R_{fi} = \frac{1}{w_k} \langle s', t' | [V_{lq}^\dagger, V_{jk}] | s, t \rangle \quad (45a)$$

$$R_{fi} = \frac{f_0^2}{2(2\pi)^3 \mu_0^2} \frac{v(q^2)v(k^2)}{\sqrt{w_q w_k^3}} \langle s', t | (\sigma \cdot q)(\sigma \cdot k) \tau_l \tau_j - (\sigma \cdot k)(\sigma \cdot q) \tau_j \tau_l | s, t \rangle \quad (45b)$$

The energy condition $E_i = E_f$, in the no recoil case translates to $k^2 = q^2$ thus

$$R_{fi} = \frac{f_0^2}{2(2\pi)^3 \mu_0^2} \frac{v(k^2)^2}{w_k^2} \langle s', t | (\sigma \cdot q)(\sigma \cdot k) \tau_l \tau_j - (\sigma \cdot k)(\sigma \cdot q) \tau_j \tau_l | s, t \rangle \quad (46)$$

Using the identities :

$$(\sigma \cdot q)(\sigma \cdot k) = i(q \times k) \cdot \sigma + k \cdot q \quad (47a)$$

$$\tau_l \tau_j = \delta_{lj} + i\epsilon_{ljm} \tau^m \quad (47b)$$

and focusing on the process $P + \pi^0 \rightarrow P + \pi^0$ we get

$$R_{fi} = \frac{if_0^2}{(2\pi)^3 \mu_0^2} \frac{v(k^2)^2}{w_k^2} \langle s' | (q \times k) \cdot \sigma | s \rangle \quad (48)$$

Next if we consider the transition amplitude square, and sum over final and averaging over initial spin states we get :

$$|\bar{R}_{fi}^2| = \left\{ \frac{if_0^2}{(2\pi)^3 \mu_0^2} \frac{v(k^2)^2}{w_k^2} \right\}^2 (q \times k)^2 \quad (49)$$

Next we analyze the same process while taking recoil into account. The conservation of energy leads to

$$w_q + \frac{q^2}{2M_0} = w_k + \frac{k^2}{2M_0} = E \quad (50)$$

The conservation of energy in this case also imposes $k^2 = q^2$. Using the identities 47b, we find that the transition matrix for the process $P + \pi^0 \rightarrow P + \pi^0$ is given by

$$R_{fi} = \frac{f_0^2}{2(2\pi)^3 \mu_0^2} \frac{v(k^2)^2}{w_k} \langle s' | \left(\frac{1}{D_1} - \frac{1}{D_2} \right) q \cdot k + \left(\frac{1}{D_1} + \frac{1}{D_2} \right) i(q \times k) \cdot \sigma | s \rangle \quad (51)$$

with

$$D_1 = E + \frac{q^2}{M_0} \quad (52a)$$

$$D_2 = E + \frac{q \cdot k}{M_0} \quad (52b)$$

Considering the amplitude squared and disregarding the spin of the final and in initial states we get

$$|\bar{R}_{fi}| = \left\{ \frac{f_0^2}{2(2\pi)^3 \mu_0^2} \frac{v(k^2)^2}{w_k} \right\}^2 \left[\left(\frac{1}{D_1} - \frac{1}{D_2} \right)^2 (q \cdot k)^2 + \left(\frac{1}{D_1} + \frac{1}{D_2} \right)^2 (q \times k)^2 \right] \quad (53)$$

3 The Wigner-Smith Operator

3.1 Historical Over View

When studying scattering processes one problem that is hard to define is the interaction time: given a certain reference frame how long does a system of interaction particle takes to go from an initial state to a final state. It was Eisenbud who suggest a solution to this problem by suggestion that the derivative of the phase shift experienced by a wave due to entering an interaction region could be interpreted as the delay due to the interaction. Building on this interpretation, Wigner set a maximum value for the time delay a

wave can experience if the interaction region is to be confined within a sphere of radius R [4]. This work was essential as it helped enforce the interpretation of the phase shift as a time delay. However the concept of time delay was still ill defined because in Wigner work it still depended on the radius of the interaction, a quantity arbitrary defined. In 1960, Smith proposed a better way to think of time delay due to interaction [5]. *"If the collision lifetime is defined as the limit, as $R \rightarrow \infty$, of the difference between the time the particles spend within a distance of each other and the time they would have spent there in the absence of the interaction, a well-defined quantity emerges which is finite as long as the interaction vanishes rapidly enough at large R ."* To formalize this idea Smith proposed a time delay operator (Wigner Smith operator), constructed from the scattering matrix, whose eigenvalues are interpreted as the time delay a state experience.

$$Q = -iS^\dagger \frac{dS}{dE} \quad (54)$$

While the concept of time delay is lesser used now in scattering theory, the Wigner Smith operator found it's use in multiple other fields. For instance, in electromagnetism the operator was used to identify waves that have a define scattering time inside a system [6]. A generalized form of the operator was then introduced to find wave forms that exploits specific proprieties. Instead of deriving the scattering matrix S with respect to energy E as it's the case for the classical WS operator, the authors in [7] derive the scattering matrix with respect the radius of the scatterer this leads to wave eigen states of the WS operator that exercise a force on the scatterer.

3.2 Smith stationary time delay

In this section we retrace the derivation of the time delay, as defined by Smith in 1960, in the review article by C.A.A. de Carvalho, H.M. Nussenzveig [8]. As mentioned before Smith proposed a definition of time as the difference a particle spends near a region of R of the scattering center. The In his original work, Smith only considered stationary solution of the Schrodinger equation. First, the dwell time in a a spherical region of radius R of the scattering center, is defined as the ratio of the probility that a state is localized in this region to the inward probability flux into the region. let $|\psi\rangle$ be the state describing the system, and j the probility density vector, then the dwell time is given by :

$$t_R = \frac{\int_{|x|<R} |\psi|^2 d^3r}{\Phi_R} \quad (55)$$

with Φ_R , is the probability flux through the surface of the spherical region (See appendix for derivation of the flux form):

$$\Phi_R = - \oint j_{in} \cdot ds = - \frac{\hbar R^2}{2im} \oint \psi_{in}^* \frac{\partial \psi}{\partial r} - \psi_{in} \frac{\partial \psi_{in}^*}{\partial r} d\Omega \quad (56)$$

In potential scattering theory, we usually consider asymptotic solutions to the Schrodinger equations, looking at the numerator this poses a problem as computing the numerator, in this current form indicates the necessity of integrating over the volume of the spherical region this means that an expression for the wave function for arbitrary small distance r is needed. This difficulty could be surmounted by using Gauss theorem and changing the volume integral to a surface integral, than calculating the dwell time for sufficiently large radius R allows us to use the asymptotic expression for the wave function. To this end consider the the stationary Schrodinger equation:

$$H\psi = E\psi \quad (57)$$

Taking the derivative with respect to energy we get,

$$H \frac{\partial \psi}{\partial E} = \psi + E \frac{\partial \psi}{\partial E} \quad (58)$$

So that :

$$\psi^* \psi = \psi^* H \psi - E \psi^* \frac{\partial \psi}{\partial E} \quad (59a)$$

$$\psi^* \psi = \psi^* H \psi - (H \psi)^* \frac{\partial \psi}{\partial E} \quad (59b)$$

$$\psi^* \psi = \frac{-\hbar^2}{2m} (\psi^* \frac{\partial}{\partial E} \nabla^2 \psi - \nabla^2 \psi^* \frac{\partial \psi}{\partial E} + \nabla \psi^* \frac{\partial}{\partial E} \nabla \psi - \nabla \psi^* \frac{\partial}{\partial E} \nabla \psi) \quad (59c)$$

$$\psi^* \psi = \frac{-\hbar^2}{2m} \nabla \cdot (\psi^* \frac{\partial}{\partial E} \nabla \psi - \nabla \psi^* \frac{\partial \psi}{\partial E}) \quad (59d)$$

Using Gauss theorem we get :

$$\int_{|x|<R} |\psi|^2 d^3r = \frac{\hbar^2 R^2}{2m} \oint \left(\frac{\partial \psi}{\partial E} \frac{\partial \psi}{\partial r} - \psi^* \frac{\partial^2 \psi}{\partial r \partial E} \right)_{r=R} d\Omega \quad (60)$$

Using the stationary solution to the free Schrodinger equation in spherical coordinates:

$$\psi_{in}(r, E) = \frac{e^{-ikr}}{r} A_l(\theta, \phi) \quad (61)$$

with

$$k = \sqrt{\frac{2mE}{\hbar^2}} \quad (62)$$

We can evaluate the flux of probability to the spherical region surrounding the scattering center,

$$\Phi_R = v B_E \quad (63)$$

with :

$$v = \frac{\hbar k}{m} \quad (64a)$$

$$B_E = \int |A_l(\theta, \Phi)|^2 d\Omega \quad (64b)$$

Consider next the stationary Schrodinger equation with the potential V :

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V(r) \psi = E \psi \quad (65)$$

If we assume that the potential decays quickly enough then asymptotically, we retrieve the free Schrodinger equation and thus the effect of the potential is restricted to change in the phase of the asymptotic wave function. Thus for a given value of the wave's angular momentum we get :

$$\psi(r, \theta, \phi) = 2i(-1)^l \frac{A_l(\theta, \phi)}{r} e^{i\eta_l(k)} \sin(kr + \eta_l(k) + \frac{\pi l}{2}) \quad (66)$$

We therefore can calculate the probability of at a time t the state be localized inside the sphere of radius R around the scattering center,

$$\int_{|r|<R} |\psi|^2 d^3r = 2B_l(R + \frac{d\eta_l}{dk} - \frac{(-1)^l}{2k} \sin(2kr + 2\eta_l(k)) + O(\frac{1}{R})) \quad (67)$$

Applying now the definition of the dwell time 55 for state with angular momentum l,

$$t_{R,l} = \frac{2R}{v} + \frac{2}{v} \frac{d\eta_l}{dk} - \frac{(-1)^l}{kv} \sin(2(kr + \eta_l)) + O(\frac{1}{R}) \quad (68)$$

The last term in the expression of the dwell time of an interacting states oscillatory and it links the uncertainty principle [9]. if we take the average of the dwell time given by :

$$\langle t_{R,l} \rangle = \frac{1}{R} \int_R^{2R} t_{R,l} dR \quad (69)$$

Then we can the oscillatory term vanishes.

For a free propagating particle (in absence of the scattering potential) a similar calculation leads :

$$\langle t_{R,l}^0 \rangle = \frac{2R}{v} \quad (70)$$

Which is in agreement with the classical time it takes an unperturbed object to traverse a sphere of radius R.

Finally Smith defines the time delay a state experience due to the presence of a scattering potential as :

$$\Delta t_l = \lim_{R \rightarrow +\infty} [\langle t_{R,l} \rangle - \langle t_{R,l}^0 \rangle] \quad (71)$$

. In this case we get :

$$\Delta t_l = 2\hbar \frac{d\eta_l}{dk} \quad (72)$$

This indeed the same as Eisenbud-Wigner partial wave time delay.

Next we try to compute the derivative of the phase shift. Let ψ be the wave function of the system. If we note :

$$\psi_l(r, \theta, \phi) = A_l(\theta, \phi) \frac{u_l(E, r)}{r} \quad (73)$$

the $u_l(r)$ satisfies the following equation :

$$-\frac{\partial^2 u_l}{\partial r^2} + \left[\frac{l(l+1)}{r^2} + \frac{2m}{\hbar} V(r) \right] u_l = k^2 u_l \quad (74)$$

Consider next solution that satisfies

$$u_l(E, 0) = 0 \quad (75a)$$

$$u_l(E, r) \rightarrow w_l(E, r) = \sin\left(kr + \frac{\eta_l}{2} - \frac{l\pi}{2}\right) (r \rightarrow \infty) \quad (75b)$$

Consider next two solution for the radial equation for 2 different energies $u_l(E_1, r)$ and $u_l(E_2, r)$ which we note respectively as $u_{l,1}$ and $u_{l,2}$. Then we get

$$\frac{\partial}{\partial r} \left(u_{l,1} \frac{\partial u_{l,2}}{\partial r} - \frac{\partial u_{l,1}}{\partial r} u_{l,2} \right) = u_{l,1} \frac{\partial^2 u_{l,2}}{\partial r^2} - \frac{\partial^2 u_{l,1}}{\partial r^2} u_{l,2} \quad (76a)$$

$$\frac{\partial}{\partial r} \left(u_{l,1} \frac{\partial u_{l,2}}{\partial r} - \frac{\partial u_{l,1}}{\partial r} u_{l,2} \right) = (k_1^2 - k_2^2) u_{l,1} u_{l,2} \quad (76b)$$

where in going from 76a to 76b we used equation 74. The same calculations for the corresponding $w_{l,1}$ and $w_{l,2}$ leads to

$$\frac{\partial}{\partial r} \left(w_{l,1} \frac{\partial w_{l,2}}{\partial r} - \frac{\partial w_{l,1}}{\partial r} w_{l,2} \right) = (k_1^2 - k_2^2) w_{l,1} w_{l,2} \quad (77)$$

Next consider :

$$J = \int_0^{+\infty} (u_{l,1} u_{l,2} - w_{l,1} w_{l,1}) dr \quad (78)$$

Using the 2 previous expression we get

$$J = \left[\frac{u_{l,1} \frac{\partial u_{l,2}}{\partial r} - \frac{\partial u_{l,1}}{\partial r} u_{l,2}}{k_1^2 - k_2^2} - \frac{w_{l,1} \frac{\partial w_{l,2}}{\partial r} - \frac{\partial w_{l,1}}{\partial r} w_{l,2}}{k_1^2 - k_2^2} \right]_0^{+\infty} \quad (79a)$$

$$J = w_{l,1}(0) \frac{\partial w_{l,2}}{\partial r}(0) - \frac{\partial w_{l,1}}{\partial r}(0) w_{l,2}(0) \quad (79b)$$

Where in going from 79a to 79b the boundary conditions 75 have been used. The first term in J is

$$J_1 = \frac{k_2 \cos(\eta_l(k_2) - \frac{\pi l}{2}) \sin(\eta_l(k_1) - \frac{\pi l}{2})}{k_1^2 - k_2^2} \quad (80)$$

Taking the limit as k_1 approaches k_2 using the l'Hopital rule we get:

$$\lim_{k_1 \rightarrow k_2} \frac{\sin(\eta_l(k_1) - \frac{\pi l}{2})}{k_1^2 - k_2^2} = \lim_{k_1 \rightarrow k_2} \frac{\frac{d\eta_l}{dk} \cos(\eta_l(k_1) - \frac{\pi l}{2})}{2k_1} \quad (81a)$$

$$\lim_{k_1 \rightarrow k_2} \frac{\sin(\eta_l(k_1) - \frac{\pi l}{2})}{k_1^2 - k_2^2} = \frac{1}{2k_2} \frac{d\eta_l}{dk}(k_2) \cos(\eta_l(k_2) - \frac{\pi l}{2}) \quad (81b)$$

Thus we get

$$\lim_{k_1 \rightarrow k_2} J_1 = \frac{1}{2} \frac{d\eta_l}{dk}(k_2) \cos(\eta_l(k_2) - \frac{\pi l}{2})^2 \quad (82)$$

A similar calculation for the second term in the expression of J leads to

$$\lim_{k_1 \rightarrow k_2} J_2 = \lim_{k_1 \rightarrow k_2} \frac{k_1 \cos(\eta_l(k_1) - \frac{\pi l}{2}) \sin(\eta_l(k_2) - \frac{\pi l}{2})}{k_1^2 - k_2^2} \quad (83a)$$

$$\lim_{k_1 \rightarrow k_2} J_2 = \frac{(-1)^l}{4k_2} \sin(2\eta_l(k_2)) - \frac{1}{2} \frac{d\eta_l}{dk}(k_2) \sin(\eta_l(k_2) - \frac{\pi l}{2})^2 \quad (83b)$$

Finally

$$\lim_{k_1 \rightarrow k_2} J = \lim_{k_1 \rightarrow k_2} J_1 - \lim_{k_1 \rightarrow k_2} J_2 \quad (84a)$$

$$\lim_{k_1 \rightarrow k_2} J = \frac{1}{2} \frac{d\eta_l}{dk}(k_2) - \frac{(-1)^l}{4k_2} \sin(2\eta_l(k_2)) \quad (84b)$$

Going back to the expression of J given by equation 75, and taking the limit as k_1 approaches k_2 we get:

$$\lim_{k_1 \rightarrow k_2} J = \int_0^r (u_{l,2}^2 - w_{l,2}^2) dr \quad (85)$$

Thus we get :

$$\frac{d\eta_l}{dk} = (-1)^l \frac{\sin(2\eta_l(k))}{2k} + 2 \int_0^{+\infty} (u_l(E, r)^2 - w_l(E, r)^2) dr \quad (86)$$

Hence the time delay is given by

$$\Delta t_l = (-1)^l \frac{\sin(2\eta_l(k)) \hbar}{k} + 4\hbar \int_0^{+\infty} (u_l(E, r)^2 - w_l(E, r)^2) dr \quad (87)$$

3.3 The Time Delay Operator

The aim of this section is to give some insights on the derivation of the Wigner Smith operator, this section is again based on the review article [8] and more rigorous prove of the form of the Wigner Smith operator can be found in [10].

Consider a scattering system characterized by its free propagation Hamiltonian H_0 and the interaction Hamiltonian H . Next we define the free and interacting time evolution operators:

$$U_0(t) = \exp(-iH_0 t) \quad (88a)$$

$$U(t) = \exp(-iH t) \quad (88b)$$

$$(88c)$$

We can define the Møller operators :

$$\Omega_{\pm} = \lim_{t \rightarrow \mp \infty} U^{\dagger}(t) U_0(t) \quad (89)$$

The Møller operator effects could be understood as following: When studying an interacting system of particles it's usually assumed that there excised a moment in the past where the different parts of the system where sufficiently far from each other that no interaction took place. Considering that in this moment no interaction took place the state of the system is described in terms of the non interacting Hamiltonian, in other words if the system at time t is described by a state $|\psi(t)\rangle$,

$$\lim_{t \rightarrow -\infty} |\psi(t)\rangle = |\phi_i\rangle \quad (90)$$

Similarly, if one is studying a scattering process, than the state of the system in the far future (away from the interaction region) is again described by the non interacting Hamiltonian. Thus, we get

$$\lim_{t \rightarrow +\infty} |\psi(t)\rangle = |\phi_f\rangle \quad (91)$$

If then suppose that state space of the interacting system, could be decomposed in 2 sets of complete eigen states: incoming and outgoing, The Møller operator Ω_- then link an eigen states of the non interacting system to the corresponding outgoing eigen state of the interacting system. Similarly, the operator Ω_+ links

a, eigen state of the non interacting system to the corresponding incoming eigen states of the interacting one.

This becomes clearer, if we use the following expressions :

$$\Omega_{\pm} = \sum |\psi_{\pm}^{\alpha}\rangle \langle \phi^{\alpha}| \quad (92)$$

where $|\psi_{\pm}^{\alpha}\rangle$ are the incoming(+) and outgoing (-) complete sets of eigen states of the interacting systems, and $|\phi^{\alpha}\rangle$ are the eigen states of non interacting systems it is not guaranteed that for a given scattering system the MØller operators exist. it is quite possible that the limit is equation It's important to note that the limit in the definition 89 does not exist.

Let $|\psi(t)\rangle$ is state of the system as a time t , and let Σ be a region of space constrained within a radius R of the scattering center. let $\Phi_R(t)$ be the flux of probability to this region. We normalize the state such that :

$$\lim_{R \rightarrow +\infty} \int_{-\infty}^{+\infty} \Phi_R(t) dt = 1 \quad (93)$$

Next consider P_{Σ} the projection operator to the region Σ :

$$P_{\Sigma} = \int_{|x| < R} |x\rangle \langle x| d^3x \quad (94)$$

then the probability at time t to find the state in the region Σ is given by

$$p(t) = \langle \psi(t) | P_{\Sigma} | \psi(t) \rangle \quad (95)$$

The state $\psi(t)$ verifies :

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle \quad (96a)$$

$$|\psi(0)\rangle = \Omega_- |\phi\rangle \quad (96b)$$

$$|\psi(t)\rangle = U(t) \Omega_- |\phi\rangle \quad (96c)$$

This follows because the state $|\psi(t)\rangle$ could be thought of at the time evolution of an incoming state $|\psi(0)\rangle$ which according to our previous remarks is deduced from the tates of the free Hamiltonian via the MØller operator. Using this result we get

$$p(t) = \langle \phi | \Omega_-^{\dagger} U^{\dagger}(t) P_{\Sigma} U(t) \Omega_- | \phi \rangle \quad (97)$$

Using the Intertwining relation :

$$\Omega_{\pm} U = U_0 \Omega_{\pm} \quad (98)$$

We get

$$p(t) = \langle \phi | U_0^{\dagger}(t) \Omega_-^{\dagger} P_{\Sigma} \Omega_- U_0(t) | \Phi \rangle \quad (99)$$

The dwell time is then defined as the integral over time of the $p(t)$

$$T_R = \int_{-\infty}^{+\infty} p(t) dt = \int_{-\infty}^{+\infty} \langle \phi | U_0^{\dagger}(t) \Omega_-^{\dagger} P_{\Sigma} \Omega_- U_0(t) | \Phi \rangle dt \quad (100)$$

For the free propagating particle we get:

$$T_{0R} = \int_{-\infty}^{+\infty} \langle \phi | U_0^{\dagger}(t) P_{\Sigma} U_0(t) | \Phi \rangle dt \quad (101)$$

and the time delay is therefore given as :

$$\tau = \lim_{R \rightarrow +\infty} T_R - T_{R0} = \lim_{R \rightarrow +\infty} \int_{-\infty}^{+\infty} \langle \phi | U_0^{\dagger}(t) (\Omega_-^{\dagger} P_{\Sigma} \Omega_- - P_{\Sigma}) U_0(t) | \Phi \rangle dt \quad (102)$$

This leads to the definition of the time delay operator by it's expectation values for given states

$$\langle \psi | Q | \psi \rangle = \lim_{R \rightarrow +\infty} \int_{-\infty}^{+\infty} \langle \phi | U_0^{\dagger}(t) (\Omega_-^{\dagger} P_{\Sigma} \Omega_- - P_{\Sigma}) U_0(t) | \Phi \rangle dt \quad (103)$$

4 Application of the Wigner-Smith operator to the Chew-Low model

The aim of the following work is to derive the W_S operator in the context of low energy meson nucleon theory as described by the Chew-Low model. Working in the context of second order perturbation theory and given an initial and final states $|\Phi_i\rangle$ and $|\Phi_f\rangle$ the scattering matrix element S_{fi} is given by :

$$S_{fi} = \langle \Phi_f | S | \Phi_i \rangle = \delta_{\Phi_i \Phi_f} - 2\pi i \delta(E_f - E_i) [\langle \Phi_f | H_I | \Phi_i \rangle + \langle \Phi_f | H_I \frac{1}{E_i - H_0 + i\epsilon} H_I | \Phi_i \rangle] \quad (104)$$

Next Assume that the energy conservation is respected i.e, $E_f = E_i$, taking the derivative over E_i than leads to

$$\frac{dS_{fi}}{dE} = 2\pi i \langle \Phi_f | H_I \frac{1}{(E_i - H_0 + i\epsilon)^2} H_I | \Phi_i \rangle \quad (105)$$

Next assuming that Φ_i and Φ_f are part of a complete set of states Φ_j we can compute to second order the WS operator :

$$\langle \Phi_f | Q | \Phi_i \rangle = -i \sum_{\Phi_j} \langle \Phi_f | S^\dagger | \Phi_j \rangle \langle \Phi_j | \frac{dS}{dE} | \Phi_i \rangle \quad (106)$$

Given that $\frac{dS}{dE}$ is already in second order in the coupling constant only the zeroth term of $\langle \Phi_f | S^\dagger | \Phi_j \rangle$ participate to the second order term of Q , thus we get

$$\langle \Phi_f | Q | \Phi_i \rangle = -i \sum_{\Phi_j} \delta_{\Phi_j \Phi_f} 2\pi i \langle \Phi_j | H_I \frac{1}{(E_i - H_0 + i\epsilon)^2} H_I | \Phi_i \rangle \quad (107a)$$

$$\langle \Phi_f | Q | \Phi_i \rangle = 2\pi \langle \Phi_j | H_I \frac{1}{(E_i - H_0 + i\epsilon)^2} H_I | \Phi_i \rangle \quad (107b)$$

Finlay reinforcing the energy conservation condition we get the final expression for the matrix element Q_{fi}

$$\langle \Phi_f | Q | \Phi_i \rangle = 2\pi \delta(E_f - E_i) \langle \Phi_j | H_I \frac{1}{(E_i - H_0 + i\epsilon)^2} H_I | \Phi_i \rangle \quad (108)$$

Returning to the Chew-Low model, we try to find an eigen state of the WS operator restricted to the on meson one nucleon state. To this end let

$$|\psi\rangle = \sum_{j,s,t} \int d^3k C_{j,s,t}(k) |j, s, t, k\rangle \quad (109)$$

be an eigen state of Q with an eigen value λ . using the eigen state equation :

$$Q |\psi\rangle = \lambda |\psi\rangle \quad (110)$$

we get

$$\langle j_1, s_1, t_1, k_1 | Q | \psi \rangle = \lambda C_{j_1, s_1, t_1}(k_1) \quad (111)$$

However using the expression of Q in terms of the interaction Hamiltonian [107b](#). The same inner product could be written as

$$\langle j_1, s_1, t_1, k_1 | Q | \psi \rangle = \sum_{j,s,t} \int d^3k C_{j,s,t}(k) \langle j_1, s_1, t_1, k_1 | Q | j, s, t, k \rangle \quad (112a)$$

$$\lambda C_{j_1, s_1, t_1}(k_1) = 2\pi \sum_{j,s,t} \int d^3k C_{j,s,t}(k) \delta(E_f - E_i) \langle j_1, s_1, t_1, k_1 | H_I \frac{1}{(E_i - H_0 + i\epsilon)^2} H_I | j, s, t, k \rangle \quad (112b)$$

Using the same techniques as in the previous section, equation [112](#) could be written as

$$(\lambda - 2\pi \hat{L}(k_1)) C_{j_1, s_1, t_1}(k_1) = 2\pi \sum_{j,s,t} \int d^3k C_{j,s,t}(k) \delta(E_k - E_{k_1}) \left(\frac{\langle s, t_1 | V_{jk} V_{j_1 k_1}^\dagger | s, t \rangle}{(E_k - E_{k_1} - f(k, k_1))^2} + \frac{\langle s, t_1 | V_{j_1 k_1}^\dagger V_{jk} | s, t \rangle}{(E_k - E_{k_1} f(k_1, k_1))^2} \right) \quad (113)$$

With :

$$\hat{L}(k_1) = \int d^3p \frac{3f^2}{2(2\pi)^3 \mu_0^2} \frac{v(p^2)}{w_p} \frac{1}{f(p, k_1)^2} \quad (114)$$

Next we focus on the No-recoil case for which :

$$f(k, k_1) = w_k; \quad (115a)$$

$$f(k_1, k_1) = w_{k_1} \quad (115b)$$

$$E_k = w_k \quad (115c)$$

$$\hat{L}(k_1) = \hat{L} = \text{constant} \quad (115d)$$

Equation 113 simplifies :

$$(\lambda - 2\pi \hat{L}(k_1)) C_{j_1, s_1, t_1}(k_1) = 2\pi \sum_{j, s, t} \int d^3k C_{j, s, t}(k) \delta(w_k - w_{k_1}) \left(\frac{\langle s, t_1 | V_{jk} V_{j_1 k_1}^\dagger | s, t \rangle}{w_k^2} + \frac{\langle s, t_1 | V_{j_1 k_1}^\dagger V_{jk} | s, t \rangle}{w_{k_1}^2} \right) \quad (116)$$

Recoiling that :

$$V_{jk} V_{j_1 k_1}^\dagger = \frac{f^2}{2(2\pi)^3 \mu_0^2} \frac{v(k^2) v(k_1^2)}{\sqrt{w_k w_{k_1}}} (k \cdot \sigma)(k_1 \cdot \sigma) \tau_j \tau_{j_1} \quad (117)$$

To go further, we consider a refrance frame where k_1 is along the z axis and we use spherical coordinates such that

$$k = \begin{pmatrix} k \sin(\theta) \cos(\phi) \\ k \sin(\theta) \sin(\phi) \\ k \cos(\theta) \end{pmatrix} \quad (118)$$

such that :

$$k \cdot \sigma = k \sigma_r(\theta, \phi) \quad (119)$$

with

$$\sigma_r(\theta, \phi) = \begin{pmatrix} \cos(\theta) & \sin(\theta) e^{i\phi} \\ \sin(\theta) e^{-i\phi} & -\cos(\theta) \end{pmatrix} \quad (120)$$

Next we define

$$\sigma_R(\theta, \phi) = \sigma_r(\theta, \phi) \cdot \sigma_r(0, 0) \quad (121a)$$

$$\sigma_L(\theta, \phi) = \sigma_r(0, 0) \cdot \sigma_r(\theta, \phi) \quad (121b)$$

Finally we decompose the wight functions into a radial and angular part :

$$C_{j, s, t}(k) = C_{j, s, t}(k, \theta, \phi) \quad (122)$$

With this definitions we can rewrite equation 113 as :

$$\begin{aligned} (\lambda - 2\pi \hat{L}) C_{j_1, s_1, t_1}(k, 0, 0) &= \frac{f^2}{8\pi^2 \mu_0^2} \sum_{j, s, t} \int_0^{2\pi} d\phi \int_0^\pi d\theta \int_0^{+\infty} dk \\ &\frac{k^3 k_1}{\sqrt{w_k w_{k_1}}} v(k^2) v(k_1^2) \frac{1}{w_k^2} \delta(w_k - w_{k_1}) \sin(\theta) C_{j, s, t}(k, \theta, \phi) \langle t_1 | \tau_j \tau_{j_1} | t \rangle \langle s_1 | \sigma_R(\theta, \phi) | s \rangle \\ &+ \frac{k^3 k_1}{\sqrt{w_k w_{k_1}}} v(k^2) v(k_1^2) \frac{1}{w_{k_1}^2} \delta(w_k - w_{k_1}) \sin(\theta) C_{j, s, t}(k, \theta, \phi) \langle t_1 | \tau_{j_1} \tau_j | t \rangle \langle s_1 | \sigma_L(\theta, \phi) | s \rangle \end{aligned} \quad (123)$$

Focus now on the radial part of equation 123 :

$$I_1 = \int_0^{+\infty} dk \frac{k^3 k_1}{\sqrt{w_k w_{k_1}}} v(k^2) v(k_1^2) \frac{C_{j, s, t}(k, \theta, \phi)}{w_k^2} \delta(w_k - w_{k_1}) \quad (124)$$

Recoiling :

$$w_k = \sqrt{k^2 + \mu_0^2} \quad (125)$$

we get

$$\frac{dw_k}{dk} = \frac{k}{w_k} \quad (126)$$

Let

$$\hat{C}_{j,s,t}(w_k, \theta, \phi) = C_{j,s,t}(k, \theta, \phi) \quad (127a)$$

$$\hat{v}(w_k) = v(k) \quad (127b)$$

changing the integration variable from k to w_k we get

$$I_1 = \int_{\mu_0}^{+\infty} dw_k \frac{(w_k^2 - \mu_0^2) \sqrt{w_{k_1}^2 - \mu_0^2}}{\sqrt{w_k^3 w_{k_1}}} \hat{C}_{j,s,t}(w_k, \theta, \phi) \hat{v}(w_k) \hat{v}(w_{k_1}) \delta(w_k - w_{k_1}) \quad (128a)$$

$$I_1 = \frac{(w_{k_1}^2 - \mu_0^2)^{\frac{3}{2}}}{w_{k_1}^2} \hat{v}(w_{k_1})^2 \hat{C}_{j,s,t}(w_{k_1}, \theta, \phi) \quad (128b)$$

With this equation 123 becomes

$$(\lambda - 2\pi \hat{L}) \hat{C}_{j_1, s_1, t_1}(w_{k_1}, 0, 0) = \frac{f^2}{8\pi^2 \mu_0^2} \frac{(w_{k_1}^2 - \mu_0^2)^{\frac{3}{2}}}{w_{k_1}^2} \hat{v}(w_{k_1})^2 \sum_{j,s,t} \int_0^{2\pi} d\phi \int_0^\pi d\theta \quad (129)$$

$$\sin(\theta) \hat{C}_{j,s,t}(w_k, \theta, \phi) \langle t_1 | \tau_j \tau_{j_1} | t \rangle \langle s_1 | \sigma_R(\theta, \phi) | s \rangle + \sin(\theta) \hat{C}_{j,s,t}(w_k, \theta, \phi) \langle t_1 | \tau_{j_1} \tau_j | t \rangle \langle s_1 | \sigma_L(\theta, \phi) | s \rangle$$

To advance further we make the observation that :

$$\sigma_R(\theta, \phi) = \begin{pmatrix} \cos(\theta) & -\sin(\theta)e^{i\phi} \\ \sin(\theta)e^{-i\phi} & \cos(\theta) \end{pmatrix} \quad (130a)$$

$$\sigma_L(\theta, \phi) = \begin{pmatrix} \cos(\theta) & \sin(\theta)e^{i\phi} \\ -\sin(\theta)e^{-i\phi} & \cos(\theta) \end{pmatrix} \quad (130b)$$

So that $\sigma_L(\pi - \theta, \phi) = -\sigma_R(\theta, \phi)$. If we now suppose that $\hat{C}_{j,s,t}(w_k, \pi - \theta, \phi) = C_{j,s,t}(w_k, \theta, \phi)$, equation 123 becomes :

$$\begin{aligned} (\lambda - 2\pi \hat{L}) \hat{C}_{j_1, s_1, t_1}(w_{k_1}, 0, 0) &= \frac{f^2}{8\pi^2 \mu_0^2} \frac{(w_{k_1}^2 - \mu_0^2)^{\frac{3}{2}}}{w_{k_1}^2} \hat{v}(w_{k_1})^2 \sum_{j,s,t} \langle t_1 | \tau_j \tau_{j_1} - \tau_{j_1} \tau_j | t \rangle \\ &\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) \hat{C}_{j,s,t}(w_k, \theta, \phi) \langle s_1 | \sigma_R(\theta, \phi) | s \rangle \end{aligned} \quad (131)$$

Using the commutation rules for Pauli :

$$\tau_j \tau_{j_1} - \tau_{j_1} \tau_j = 2\epsilon_{jj_1 k} \tau^k \quad (132)$$

we finally get :

$$\begin{aligned} (\lambda - 2\pi \hat{L}) \hat{C}_{j_1, s_1, t_1}(w_{k_1}, 0, 0) &= \frac{f^2}{4\pi^2 \mu_0^2} \frac{(w_{k_1}^2 - \mu_0^2)^{\frac{3}{2}}}{w_{k_1}^2} \hat{v}(w_{k_1})^2 \sum_{j,s,t} \epsilon_{jj_1 k} \langle t_1 | \tau^k | t \rangle \\ &\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) \hat{C}_{j,s,t}(w_k, \theta, \phi) \langle s_1 | \sigma_R(\theta, \phi) | s \rangle \end{aligned} \quad (133)$$

Another possible choice of the transformation proprieties of $\hat{C}_{j,s,t}(w_k, \theta, \phi)$ is $\hat{C}_{j,s,t}(w_k, \pi - \theta, \phi) = -C_{j,s,t}(w_k, \theta, \phi)$, this actually simplifies the equations further because we get :

$$\begin{aligned} (\lambda - 2\pi \hat{L}) \hat{C}_{j_1, s_1, t_1}(w_{k_1}, 0, 0) &= \frac{f^2}{8\pi^2 \mu_0^2} \frac{(w_{k_1}^2 - \mu_0^2)^{\frac{3}{2}}}{w_{k_1}^2} \hat{v}(w_{k_1})^2 \sum_{j,s,t} \langle t_1 | \tau_j \tau_{j_1} + \tau_{j_1} \tau_j | t \rangle \\ &\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) \hat{C}_{j,s,t}(w_k, \theta, \phi) \langle s_1 | \sigma_R(\theta, \phi) | s \rangle \end{aligned} \quad (134)$$

Then using the aniti-commutation rules of the Pauli matrices :

$$\tau_j \tau_{j_1} + \tau_{j_1} \tau_j = 2\delta_{j,j_1} \quad (135)$$

we get

$$\begin{aligned} (\lambda - 2\pi\hat{L})\hat{C}_{j_1,s_1,t_1}(w_{k_1}0,0) &= \frac{f^2}{4\pi^2\mu_0^2} \frac{(w_{k_1}^2 - \mu_0^2)^{\frac{3}{2}}}{w_{k_1}^2} \hat{v}(w_{k_1})^2 \\ \sum_s \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) \hat{C}_{j_1,s,t_1}(w_k\theta,\phi) \langle s_1 | \sigma_R(\theta,\phi) | s \rangle \end{aligned} \quad (136)$$

4.1 Analytical solution for equation 136

Given that equation 136 suggest a no dependence on the isospin variables, we propose to omit this variables in the rest of this paragraph.

$$(\lambda - 2\pi\hat{L})\hat{C}_{s_1}(w_{k_1},0,0) = \frac{f^2}{4\pi^2\mu_0^2} \frac{(w_{k_1}^2 - \mu_0^2)^{\frac{3}{2}}}{w_{k_1}^2} \hat{v}(w_{k_1})^2 \sum_s \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) \hat{C}_s(w_k,\theta,\phi) \langle s_1 | \sigma_R(\theta,\phi) | s \rangle \quad (137)$$

Next let :

$$M(w_{k_1},\lambda) = \frac{1}{4}(\lambda - 2\pi\hat{L}) \left(\frac{f^2}{4\pi^2\mu_0^2} \frac{(w_{k_1}^2 - \mu_0^2)^{\frac{3}{2}}}{w_{k_1}^2} \hat{v}(w_{k_1})^2 \right)^{-1} \quad (138)$$

if we next suppose also that $\hat{C}_s(w_k,\theta,2\pi+\phi) = -\hat{C}_s(w_k,\theta,\phi)$ we can rewrite the left part as :

$$4M(w_{k_1},\lambda)\hat{C}_{s_1}(w_{k_1},0,0) = M(w_k,\lambda) \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\partial^2 \hat{C}_s(w_{k_1},\theta,\phi)}{\partial\theta\partial\phi} \quad (139)$$

The integral equation 137 could then be rewritten as a differential equation :

$$M(w_{k_1},\lambda) \frac{\partial^2 \hat{C}_{s_1}(w_{k_1},\theta,\phi)}{\partial\theta\partial\phi} = \sum_s \sin(\theta) \hat{C}_s(w_{k_1}\theta,\phi) \langle s_1 | \sigma_R(\theta,\phi) | s \rangle \quad (140)$$

Focus next on the spin term $\langle s_1 | \sigma_R(\theta,\phi) | s \rangle$. If we represent the spin $\frac{1}{2}$ state with column vector $(1,0)^T$ and spin $-\frac{1}{2}$ with the vector $(0,-1)^T$ then

$$\langle s_1 | \sigma_R(\theta,\phi) | s \rangle = (s_1 + \frac{1}{2})[\cos\theta(s + \frac{1}{2}) - \sin\theta e^{i\phi}(\frac{1}{2} - s)] + (\frac{1}{2} - s_1)[\sin\theta e^{-i\phi}(s + \frac{1}{2}) + \cos\theta(\frac{1}{2} - s_1)] \quad (141)$$

Using this expression we can rewrite equation 140 as a set of coupled differential equation in the with the functions $\hat{C}_1(w_k\theta,\phi)$ and $\hat{C}_{-1}(w_k,\theta,\phi)$ for the spin up and down :

$$M(w_k,\lambda) \frac{\partial^2 \hat{C}_1(w_k,\theta,\phi)}{\partial\theta\partial\phi} = \frac{1}{2} \sin(2\theta) \hat{C}_1(w_k,\theta,\phi) - \sin\theta^2 e^{i\phi} \hat{C}_{-1}(w_k,\theta,\phi) \quad (142a)$$

$$M(w_k,\lambda) \frac{\partial^2 \hat{C}_{-1}(w_k,\theta,\phi)}{\partial\theta\partial\phi} = \sin\theta^2 e^{-i\phi} \hat{C}_1(w_k,\theta,\phi) + \frac{1}{2} \sin(2\theta) \hat{C}_{-1}(w_k,\theta,\phi) \quad (142b)$$

If we next define :

$$C(w_k,\theta,\phi) = \begin{pmatrix} \hat{C}_1(w_k\theta,\phi) \\ \hat{C}_{-1}(w_k,\theta,\phi) \end{pmatrix} \quad (143a)$$

$$A(\theta,\phi) = \begin{pmatrix} \frac{1}{2} \sin(2\theta) & -\sin\theta^2 e^{i\phi} \\ \sin\theta^2 e^{-i\phi} & \frac{1}{2} \sin(2\theta) \end{pmatrix} \quad (143b)$$

equations 142 could be combined into one 2 dimensional deferential equation :

$$M(w_k,\lambda) \frac{\partial^2 C(w_k,\theta,\phi)}{\partial\theta\partial\phi} = A(\theta,\phi) C(w_k,\theta,\phi) \quad (144)$$

To solve this equation we first diagonalize $A(\theta, \phi)$

$$A(\theta, \phi) = V(\phi)\Lambda(\theta)V^{-1}(\phi) \quad (145)$$

With :

$$\Lambda(\theta) = \begin{pmatrix} \frac{1}{2} \sin(2\theta) + i \sin(\theta)^2 & 0 \\ 0 & \frac{1}{2} \sin(2\theta) - i \sin(\theta)^2 \end{pmatrix} \quad (146a)$$

$$V(\phi) = \begin{pmatrix} ie^{i\phi} & -ie^{i\phi} \\ 1 & 1 \end{pmatrix} \quad (146b)$$

$$V^{-1}(\phi) = \frac{1}{2ie^{i\phi}} \begin{pmatrix} 1 & ie^{i\phi} \\ -1 & ie^{i\phi} \end{pmatrix} \quad (146c)$$

Let :

$$S(w_k, \theta, \phi) = V^{-1}(\phi)C(w_k, \theta, \phi) \quad (147)$$

Then :

$$\frac{\partial^2 S}{\partial \theta \partial \phi} = \frac{\partial V^{-1}}{\partial \phi} \frac{\partial C}{\partial \theta} + V^{-1} \frac{\partial^2 C}{\partial \theta \partial \phi} \quad (148)$$

Observing that $C = VS$ we get

$$V^{-1} \frac{\partial^2 C}{\partial \theta \partial \phi} = \frac{\partial^2 S}{\partial \theta \partial \phi} - \frac{\partial V^{-1}}{\partial \phi} V \frac{\partial S}{\partial \theta} \quad (149)$$

From the relation :

$$V^{-1}V = I \quad (150)$$

we get :

$$\frac{\partial V^{-1}}{\partial \phi} V = -V^{-1} \frac{\partial V}{\partial \phi} \quad (151)$$

So that 149 could be written as

$$V^{-1} \frac{\partial^2 C}{\partial \theta \partial \phi} = \frac{\partial^2 S}{\partial \theta \partial \phi} + V^{-1} \frac{\partial V}{\partial \phi} \frac{\partial S}{\partial \theta} \quad (152)$$

Calculating the the expression :

$$V^{-1} \frac{\partial V}{\partial \phi} = \frac{1}{2i} P \quad (153)$$

with

$$P = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \quad (154)$$

Finally we can rewrite equation 144 as

$$M(w_k, \lambda) \left(\frac{\partial^2 S}{\partial \theta \partial \phi} + \frac{1}{2i} P \frac{\partial S}{\partial \theta} \right) = \Lambda(\theta) S(w_k, \theta, \phi) \quad (155)$$

A solution to equation 155, could be obtained if we suppose that we can write

$$C(w_k, \theta, \phi) = e^{i\frac{\phi}{2}} N(w_k, \theta) \quad (156)$$

This anastaz for the solution guarantees that the solution will satisfy the needed symmetry condition:

$$C(w_k, \theta, 2\pi + \phi) = -C(w_k, \theta, \phi) \quad (157)$$

Now we can evaluate the derivative :

$$\frac{\partial^2 S}{\partial \theta \partial \phi} = \left(\frac{\partial V^{-1}}{\partial \phi} + \frac{i}{2} V^{-1} \right) \frac{\partial C}{\partial \theta} \quad (158)$$

Using equation 151, we get

$$\frac{\partial^2 S}{\partial \theta \partial \phi} = (-V^{-1} \frac{\partial V}{\partial \phi} V^{-1} + \frac{i}{2} V^{-1}) \frac{\partial C}{\partial \theta} \quad (159a)$$

$$\frac{\partial^2 S}{\partial \theta \partial \phi} = (-V^{-1} \frac{\partial V}{\partial \phi} + \frac{i}{2}) \frac{\partial S}{\partial \theta} \quad (159b)$$

$$\frac{\partial^2 S}{\partial \theta \partial \phi} = (-\frac{1}{2i} P + \frac{i}{2}) \frac{\partial S}{\partial \theta} \quad (159c)$$

Where in the last line we used equation 153. Plugging this result into equation 155 we get a simplified differential equation

$$\frac{\partial N}{\partial \theta} = -\frac{2i}{M(w_k, \lambda)} \Lambda(\theta) N(w_k, \theta) \quad (160)$$

A solution to this equation is :

$$N(w_k, \theta) = \exp(-\frac{2i}{M(w_k, \lambda)} \Lambda(\bar{\theta})) N_0(w_k) \quad (161)$$

With :

$$\bar{\Lambda}(\theta) = \int \Lambda(\theta) \quad (162a)$$

$$\bar{\Lambda}(\theta) = \begin{pmatrix} \frac{1}{4} e^{-2i\theta} + i\frac{\theta}{2} & 0 \\ 0 & \frac{1}{4} e^{2i\theta} - i\frac{\theta}{2} \end{pmatrix} \quad (162b)$$

Recalling now the symmetry condition $C(w_k, \pi - \theta, \phi) = -C(w_k, \theta, \phi)$ we try to find N_0 such that this symmetry condition is fulfilled. Such an $N_0(w_k)$ will satisfy the following eigenvalue equation :

$$\exp(\frac{2i}{M(w_k, \lambda)} (\bar{\Lambda}(\theta) - \bar{\Lambda}(\pi - \theta))) N_0(w_k) = -N_0(w_k) \quad (163)$$

However given that the matrices $\bar{\Lambda}$ are already diagonalized it's easy to see that -1 is not an eigenvalue of the matrix $\exp(\frac{2i}{M(w_k, \lambda)} (\bar{\Lambda}(\theta) - \bar{\Lambda}(\pi - \theta)))$ so that equation 163 has no solution. Furthermore, this whole derivation is build on a mathematically unsound argument: transforming the integral equation (a global operator) to a deferential one (a local operation) this imposes an unnecessary structure on the equation. So even if this approach have lead to solutions they won't probably correspond to physical states.

5 New Approach to the integral equation (26/07/2025)

Given that the previous method to solve the integral equation 113 failed to provide solution and had some mathematical problems namely transforming the integral equation into a deferential one. We propose a new approach to solve the problem. Consider equation 113 :

$$(\lambda - 2\pi \hat{L}(k_1)) C_{j_1, s_1, t_1}(k_1) = 2\pi \sum_{j, s, t} \int d^3 k C_{j, s, t}(k) \delta(E_k - E_{k_1}) \left(\frac{\langle s, t_1 | V_{jk} V_{j_1 k_1}^\dagger | s, t \rangle}{(E_k - E_{k_1} - f(k, k_1))^2} + \frac{\langle s, t_1 | V_{j_1 k_1}^\dagger V_{jk} | s, t \rangle}{(E_k - E_{k_1} + f(k_1, k_1))^2} \right) \quad (164)$$

Now consider the recoil model for which we have

$$f(p, p') = E_p + \frac{p \cdot p'}{M_0} \quad (165)$$

Thus we can re write equation 113 as

$$(\lambda - 2\pi \hat{L}(k_1)) C_{j_1, s_1, t_1}(k_1) = 2\pi \sum_{j, s, t} \int d^3 k C_{j, s, t}(k) \delta(E_k - E_{k_1}) \left(\frac{\langle s, t_1 | V_{jk} V_{j_1 k_1}^\dagger | s, t \rangle}{(E_{k_1} + \frac{k \cdot k_1}{M_0})^2} + \frac{\langle s, t_1 | V_{j_1 k_1}^\dagger V_{jk} | s, t \rangle}{(E_k + \frac{k_1^2}{M_0})^2} \right) \quad (166)$$

We switch to spherical coordinates (and labels $k_1 \rightarrow k$ and $k \rightarrow k'$)

$$\begin{aligned}
(\lambda - 2\pi L(k, \theta, \phi))C_{j,s,t}(k, \theta, \phi) &= \frac{f^2}{8\pi^2\mu_0^2} \sum_{j',s',t'} \int_0^{2\pi} d\phi' \int_0^\pi d\theta' \int_0^{+\infty} dk' \frac{k'^3}{\sqrt{w(k')w(k)}} v(k^2)v(k_1^2) \\
C_{j',s',t'}(k', \theta', \phi') \sin(\theta') \delta(E(k) - E(k')) &\left(\frac{1}{(E(k) + \frac{k \cdot k'}{M_0})^2} \langle t | \tau_{j'} \tau_j | t' \rangle \langle s | \sigma_R(\theta, \phi, \theta', \phi') | s' \rangle + \right. \\
&\left. \frac{1}{(E(k') + \frac{k^2}{M_0})^2} \langle t | \tau_j \tau_{j'} | t' \rangle \langle s | \sigma_L(\theta, \phi, \theta', \phi') | s' \rangle \right)
\end{aligned} \quad (167)$$

Where :

$$\sigma_R(\theta, \phi, \theta', \phi') = \sigma_r(\theta', \phi') \sigma_r(\theta, \phi) \quad (168a)$$

$$\sigma_L(\theta, \phi, \theta', \phi') = \sigma_r(\theta, \phi) \sigma_r(\theta', \phi') \quad (168b)$$

Next, we express the delta energy function in terms of the momentum, using the formula

$$\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{f'(x_i)} \quad (169)$$

Where x_i are the roots of f . Than applying this to the energy delta function we get

$$E(k) = \sqrt{\mu_0^2 + k^2} + \frac{k^2}{2M_0} \quad (170a)$$

$$\frac{dE}{dk} = \frac{k}{w(k)} + \frac{k}{M_0} \quad (170b)$$

$$\delta(E(k) - E(k')) = \frac{\delta(k - k')}{\frac{k}{w(k)} + \frac{k}{M_0}} \quad (170c)$$

Using this expressing and integrating over k we get :

$$\begin{aligned}
(\lambda - 2\pi L(k, \theta, \phi))C_{j,s,t}(k, \theta, \phi) &= \frac{f^2}{8\pi^2\mu_0^2} \frac{k^3}{M + w_k} v(k^2)^2 \sum_{j',s',t'} \int_0^{2\pi} d\phi' \int_0^\pi d\theta' C_{j',s',t'}(k', \theta', \phi') \sin(\theta') \\
&\left(\frac{1}{(E(k) + \frac{k^2 \cos(\gamma)}{M_0})^2} \langle t | \tau_{j'} \tau_j | t' \rangle \langle s | \sigma_R(\theta, \phi, \theta', \phi') | s' \rangle + \frac{1}{(E(k) + \frac{k^2}{M_0})^2} \langle t | \tau_j \tau_{j'} | t' \rangle \langle s | \sigma_L(\theta, \phi, \theta', \phi') | s' \rangle \right)
\end{aligned} \quad (171)$$

with

$$\cos \gamma = \sin(\theta) \sin(\theta') \cos(\phi - \phi') + \cos(\theta) \cos(\theta') \quad (172)$$

This equation is valid for any triplet of indexes (j, s, t) , thus we have 12 equation of the same type that we propose to summarize them in a matrix format.

First, Let

$$C(k, \theta, \phi) = \begin{pmatrix} C_{1, \frac{1}{2}, \frac{1}{2}}(k, \theta, \phi) \\ C_{1, \frac{1}{2}, -\frac{1}{2}}(k, \theta, \phi) \\ C_{1, -\frac{1}{2}, \frac{1}{2}}(k, \theta, \phi) \\ \dots \end{pmatrix} \quad (173)$$

Next we define the matrix

$$M_{(j,s,t)(j',s',t')}(k, \theta, \phi, \theta', \phi') = \frac{\langle t | \tau_{j'} \tau_j | t' \rangle \langle s | \sigma_R(\theta, \phi, \theta', \phi') | s' \rangle}{(E(k) + \frac{k^2 \cos(\gamma)}{M_0})^2} + \frac{\langle t | \tau_j \tau_{j'} | t' \rangle \langle s | \sigma_L(\theta, \phi, \theta', \phi') | s' \rangle}{(E(k) + \frac{k^2}{M_0})^2} \quad (174)$$

so that the 12 equations defined by 175, could be rewritten as :

$$(\lambda - 2\pi L(k, \theta, \phi))C(k, \theta, \phi) = \alpha(k) \int_0^{2\pi} d\phi' \int_0^\pi d\theta' \sin(\theta') M(k, \theta, \phi, \theta', \phi') C(k, \theta', \phi') \quad (175)$$

with :

$$\alpha(k) = \frac{f^2}{8\pi^2\mu_0^2} \frac{k^3}{M + w_k} v(k^2)^2 \quad (176)$$

Given that in meson nucleon interaction, the P wave dominates at low energies we propose to expand C in the form of :

$$C(k, \theta, \phi) = \sum_{m=-1}^{m=1} A_m(k) Y_m(\theta, \phi) \quad (177)$$

where A_m are vectors that depend only on the momentum amplitude et $Y_m(\theta, \phi)$ are the spherical harmonics for $l = 1$. Following the same logic we expand both the matrix M and the scalar function L

$$M(k, \theta, \phi, \theta', \phi') = \sum_{p, p'=-1}^1 K^{p, p'}(k) Y_p(\theta, \phi) Y_{p'}^*(\theta', \phi') \quad (178a)$$

$$L(k, \theta, \phi) = \sum_{d=-1}^1 a_d(k) Y_d(\theta, \phi) \quad (178b)$$

Using this new expansion we can than rewrite equation 175:

$$\begin{aligned} (\lambda - \sum_{d=-1}^1 a_d(k) Y_d(\theta, \phi)) \sum_{m=-1}^{m=1} A_m(k) Y_m(\theta, \phi) &= \alpha(k) \sum_{p, p'=-1}^1 \sum_{m'=-1}^{m'=1} K^{p, p'}(k) A_{m'}(k) Y_p(\theta, \phi) \\ &\quad \int_0^{2\pi} d\phi' \int_0^\pi d\theta' \sin(\theta') Y_{p'}(\theta', \phi') Y_{m'}(\theta', \phi') \end{aligned} \quad (179)$$

Using the propriety :

$$\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) Y_{lm}(\theta, \phi) Y_{l'm'}(\theta, \phi) = \delta_{mm'} \delta_{ll'} \quad (180)$$

We get :

$$(\lambda - \sum_{d=-1}^1 a_d(k) Y_d(\theta, \phi)) \sum_{m=-1}^{m=1} A_m(k) Y_m(\theta, \phi) = \alpha(k) \sum_{p, p'=-1}^1 K^{p, p'}(k) A_{p'}(k) Y_p(\theta, \phi) \quad (181)$$

Multiplying both sides by $Y_n^*(\theta, \phi) \sin(\theta)$ and integrating over the unit sphere we get

$$\lambda A_n(k) - \sum_{m, d=-1}^1 a_d(k) A_m(k) \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin(\theta) Y_n^*(\theta, \phi) Y_d(\theta, \phi) Y_m(\theta, \phi) = \alpha(k) \sum_p K^{n, p}(k) A_p(k) \quad (182)$$

Using Gaunt Formula :

$$\int_{S^2} Y_{\ell_1}^{m_1}(\theta, \phi) Y_{\ell_2}^{m_2}(\theta, \phi) Y_{\ell_3}^{m_3}(\theta, \phi) d\Omega = \sqrt{\frac{(2\ell_1+1)(2\ell_2+1)(2\ell_3+1)}{4\pi}} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \quad (183)$$

Setting $l_1 = l_2 = l_3 = 1$ the integral is then equal to zero. and we are left with

$$\lambda A_n(k) = \alpha(k) \sum_p K^{n, p}(k) A_p(k) \quad (184)$$

If we are next to define

$$A = \begin{pmatrix} A_{-1} \\ A_0 \\ A_1 \end{pmatrix} \quad (185a)$$

$$K(k) = \begin{pmatrix} K_{-1-1} & K_{-10} & K_{-11} \\ K_{0-1} & K_{00} & K_{01} \\ K_{1-1} & K_{10} & K_{11} \end{pmatrix} \quad (185b)$$

Then equation 184 becomes an eigenvalue equation of the form :

$$\frac{\lambda}{\alpha(k)} A(k) = K(k) A(k) \quad (186)$$

Going back to equation 178 and using the fact that the matrix M is hermitian (multiplying 4 Pauli matrices which is hermitian) we get

$$M(k, \theta, \phi, \theta', \phi') = \sum_{p, p'=-1}^1 K^{p, p'}(k) Y_p(\theta, \phi) Y_{p'}^*(\theta', \phi') \quad (187a)$$

$$M^\dagger(k, \theta, \phi, \theta', \phi') = \sum_{p, p'=-1}^1 K^{p, p'}(k) Y_p^*(\theta, \phi) Y_{p'}(\theta', \phi') \quad (187b)$$

$$(187c)$$

so that we get :

$$K^{p, p'} = K^{p', p^\dagger} \quad (188)$$

Which mean that the matrix K is hermitian and thus its eigenvalues are real. K is a 36×36 matrix which means we got 36 eigenvalues for each we got an eigen vectors A that defines the eigenvalue of the WS operator. There is one final remarque that makes the results we food a bit suspicious λ should be a constant time independent of k however according to equation ??, K is clearly dependent on the the momentum amplitude k , so most likely it's eigenvalues are too, no one can absorbs the dependents in $\alpha(k)$ through the cut-off $v(k^2)$ however this can only be possible if the dependence on k can be factorized from all the eigenvalues, which is quite unlikely. One can solve the dependency of λ on k , using a trick. However this only pushes the problem further and not really solve for we end up with the mathematically sound appearing arguments that does not agree so one of them should be wrong. The trick is the following :

Consider the definition of the eigen state of the WS operator given by :

$$|\psi\rangle = \sum_{j, s, t} \int d^3k C_{j, s, t}(k) |j, s, t, k\rangle \quad (189)$$

If we suppose that $|\psi\rangle$ is normalized so that

$$\langle\psi|\psi\rangle = \sum_{j, s, t} \int d^3k |C_{j, s, t}(k)|^2 = 1 \quad (190)$$

Using the expansion of C in terms of spherical harmonics this can be rewritten as

$$\sum_{m=-1}^{m=1} \int_0^{+\infty} k^2 |A_m(k)|^2 dk = \int_0^{+\infty} k^2 |A(k)|^2 dk = 1 \quad (191)$$

Now, Assuming that λ is independent of k , we reconsider equation 186. multiply both sides by $k^2 \alpha(k) A^\dagger(k)$ and integration over k we get :

$$\lambda \int_0^{+\infty} k^2 |A(k)|^2 dk = \int_0^{+\infty} k^2 \alpha(k) A^\dagger(k) K(k) A(k) dk \quad (192a)$$

$$\lambda = \int_0^{+\infty} k^2 \alpha(k) A^\dagger(k) K(k) A(k) dk \quad (192b)$$

Thus is assume that our first remark is correct, A is a an eigenvector of K . Then we get 36 eigenvectors A_i with eigenvalues $\nu_i(k)$. For each eigenvector eigen state doublet, we can define λ_i an eigenvalue of the WS operator given by :

$$\lambda_i = \int_0^{+\infty} k^2 \alpha(k) \nu_i |A_i(k)|^2 dk \quad (193)$$

While this different interpretation of equation 186 solves the dependence of λ over k , as mentioned before it still in opposition of the direct interpretation of 186, i.e that $\frac{\lambda}{\alpha(k)}$ is an eigenvalue of $K(k)$ and which we previously argued that this leads to λ most likely being dependent on k . However one can argue that equation 193 is build in the assumption that λ is independent of k and that A is an eigenvector of $K(k)$. I'm not really sure if this is correct. The main assumption made in this approach is the restriction to $l = 1$ angular momentum. While I think it's a physically motivated approximation maybe we should drop it to get more complete solution. For one thing the structure of the Kernel K and the function $L(k)$ defiantly include higher angular momentum values so maybe we should drop this assumption and try to create a full description without imposing restriction on angular momentum values.

6 Discussion

This report has detailed an investigation into the eigen states of the Wigner-Smith (WS) operator within the framework of the Chew-Low model. The primary objective was to solve the eigenvalue equation for the WS operator to identify states with a well-defined interaction time delay[cite: 448, 724]. To this end, two distinct analytical approaches were pursued.

The first attempt sought a direct solution to the eigenvalue equation in the no-recoil case by imposing specific symmetry conditions on the wavefunction. This path led to the conversion of the integral equation into a partial differential equation. However, as noted in the analysis, this conversion lacks rigorous mathematical justification, equating a non-local integral operation with a local differential one. Consequently, this approach was deemed to be built upon a “mathematically unsound foundation” and could not be expected to yield physically consistent eigen states.

A second, more rigorous method was then developed in the “New Approach”. This involved a partial wave expansion of the eigen state, focusing on the dominant P-wave ($l = 1$) component, a standard technique in scattering theory. This successfully transformed the integral equation into a matrix eigenvalue problem of the form $\lambda \mathbf{A}(k) = \alpha(k) \mathbf{K}(k) \mathbf{A}(k)$. This result, however, revealed a deep conceptual problem: for the equation to hold for all momentum values k , the eigenvalue λ must be dependent on k . This contradicts the fundamental definition of an eigenvalue as a constant, physically implying that the time delay would depend on the energy of the probing meson.

An attempt to resolve this issue by integrating over all momenta was explored. This procedure yielded a constant value for λ , but it was recognized that this calculation does not solve the true eigenvalue problem. Instead, it correctly corresponds to finding the *expectation value* of the time delay operator within the assumed P-wave state.

The collective failure of these analytical approaches to find a simple eigen state is the most significant finding of this report. It strongly suggests that a state composed of a single partial wave (like the P-wave) is not, in fact, a true eigen state of the Wigner-Smith operator in this model. The actual eigen states are likely more complex superpositions of multiple angular momentum states. This conclusion provides a powerful motivation for the proposed next steps, such as a full numerical diagonalization of the operator, or the development of a non-perturbative formulation of the S-matrix to construct the WS operator, bypassing the limitations of the current approach.

7 Next Steps

The next step to continue the investigation of the WS operator in the context of the Chew-Low model, the following steps are proposed :

1. Go back to the Scattering amplitudes calculate in context of the Chew-Low model and try to re derive them from a relativistic calculation of the scattering process and then taking the non relativistic limit. This would prove that the Chew-Low model is indeed a low energy limit of the relativistic field theory calculations.
2. Drop the assumption on the angular momentum and have a full equation.
3. Given that analytical solution are rather difficult to obtain, we propose to use numerical analysis instead relaying on transition matrix amplitude derived in this report we can calculate numerically the scattering matrix and thus have a numerical expression for WS operator matrix that can be numerically diagonalized.
4. Study the Low formalism to get a full scattering matrix and use it to develop the WS operator.
5. Generalize the WS operator to relativistic settings, by replacing the derivative of the scattering matrix wto the energy bu the derivative with respect to space like surfaces in the energy momentum space following the Tomonaga-Schwinger approach.

A Detailed Derivations

A.1 Proof of equation 11

Consider the interaction Hamiltonian :

$$H_I = \frac{f_0}{\mu_0} \sum_{i,j=1}^3 \int d^3x \rho(x) \sigma_i \frac{\partial \phi_j}{\partial x_i} \tau_j \quad (194)$$

Using the expansion of the meson operators 9 , we get

$$\frac{\partial \phi_j}{\partial x_i} = \frac{1}{(2\pi)^{\frac{3}{2}}} \int \frac{d^3k}{\sqrt{2w_k}} i k_i (a_{jk} e^{ik \cdot x} - a_{jk}^\dagger e^{-ik \cdot x}) \quad (195)$$

Thus the interaction is Hamiltonian is then :

$$H_I = \frac{if_0}{\mu_0} \sum_{i,j=1}^3 \int d^3x \int \frac{d^3k}{\sqrt{2\pi}^3 2w_k} \rho(x) \sigma_i k_i (a_{jk} e^{ik \cdot x} - a_{jk}^\dagger e^{-ik \cdot x}) \tau_j \quad (196)$$

Inverting the order of integration and suing the Fourier transform of the cut-off function 7 we get,

$$H_I = \frac{if_0}{\mu_0} \sum_{i,j=1}^3 \int \frac{d^3k}{\sqrt{2\pi}^3 2w_k} \sigma_i k_i (v(k^2) a_{jk} - v^\dagger(k^2) a_{jk}^\dagger) \tau_j \quad (197)$$

Finally, let V_{jk} be given by equation 12 and we get the expression of the interaction Hamiltonian in momentum space given by equation 11.

A.2 Proof of equation 22

Starting from the eigen value equation 19, we get :

$$H |s, t\rangle_+ = \Delta E |s, t\rangle_+ \quad (198a)$$

$$(H'_0 + H'_I) |s, t\rangle_+ = \sqrt{Z} \Delta E |s, t\rangle_+ + \delta E |X\rangle \quad (198b)$$

$$H'_0 |s, t\rangle_+ + H'_I |s, t\rangle_+ = H'_0 (\sqrt{Z} |s, t\rangle_+ + |X\rangle) - H'_0 |X\rangle + \Delta E |X\rangle \quad (198c)$$

$$H'_I |s, t\rangle_+ = (\Delta E - H'_0) |X\rangle \quad (198d)$$

The state $|X\rangle$ is in the orthogonal subspace to the one nucleon states. On this space the operator $\Delta E - H'_0$ is non singular and therefore an inverse exist. Thus, one can write :

$$|X\rangle = (1 - P_n) \frac{1}{\Delta E - H'_0} H'_I |s, t\rangle_+ \quad (199)$$

The $(1 - P_n)$ factor is added to ensure that the state $|X\rangle$ is in the orthogonal subspace to the one nucleon states. Injecting this expression into equation 20, we get equation 22.

A.3 Proof of equation 27

To prove equation 28 we follow the same technique found in *An Introduction to Relativistic Quantum Field Theory*. First

$$a_{jk} \frac{1}{\Delta E - H'_0} = \lim_{\epsilon \rightarrow 0} \frac{1}{\Delta E - H'_0 + i\epsilon} = \lim_{\epsilon \rightarrow 0} -i \int_0^{+\infty} d\lambda a_{jk} e^{i(\Delta E - H'_0 + i\epsilon)\lambda} \quad (200a)$$

$$a_{jk} \frac{1}{\Delta E - H'_0} = \lim_{\epsilon \rightarrow 0} -i \int_0^{+\infty} d\lambda e^{i(\Delta E + i\epsilon)\lambda} a_{jk} e^{-iH'_0\lambda} \quad (200b)$$

Next given that H'_0 is a hermitian operator, $e^{-iH'_0\lambda}$ is unitary. We can therefor rewrite the previous integral as

$$a_{jk} \frac{1}{\Delta E - H'_0} = \lim_{\epsilon \rightarrow 0} -i \int_0^{+\infty} d\lambda e^{i(\Delta E + i\epsilon)\lambda} e^{-iH'_0\lambda} e^{iH'_0\lambda} a_{jk} e^{-iH'_0\lambda} \quad (201a)$$

$$a_{jk} \frac{1}{\Delta E - H'_0} = \lim_{\epsilon \rightarrow 0} -i \int_0^{+\infty} d\lambda e^{i(\Delta E - H'_0 + i\epsilon)\lambda} e^{iH'_0\lambda} a_{jk} e^{-iH'_0\lambda} \quad (201b)$$

Using the identity :

$$e^{i\alpha S} A e^{-i\alpha S} = \sum_{n=0}^{+\infty} \frac{(i\alpha)^n}{n!} [^n S, A] \quad (202)$$

with

$$[^n S, A] = [S[S\dots\dots[S, A]] \text{ n times}] \quad (203)$$

we get

$$e^{iH'_0\lambda} a_{jk} e^{-iH'_0\lambda} = \sum_{n=0}^{+\infty} \frac{(i\lambda)^n}{n!} [^n H'_0, a_{jk}] \quad (204)$$

Let's now compute the commutator $[H'_0, a_{jk}]$. First recoiling that $H'_0 = H_0 + \Delta E$ and that ΔE is just a constant we get $[H'_0, a_{jk}] = [H_0, a_{jk}]$. Now recalling the expression for the free Hamiltonian :

$$H_0 = \sum_{j=1}^3 \int d^3k \left(\frac{k^2}{2M_0} + w_k \right) a_{jk}^\dagger a_{jk} + \sum_{j,l=1}^3 \int d^3k \int d^3k' \frac{k \cdot k'}{2M_0} a_{jk}^\dagger a_{lk'}^\dagger a_{jk} a_{lk'} \quad (\text{Recoil}) \quad (205a)$$

$$H_0 = \sum_{j=1}^3 \int d^3k w_k a_{jk}^\dagger a_{jk} \quad (\text{No-recoil}) \quad (205b)$$

A straightforward calculation then proves that

$$[H_0, a_{jk}] = -\hat{f}(k) a_{jk} \quad (206)$$

with $\hat{f}(k)$ is given by equations 28. A simple iteration than proves that :

$$[^n H'_0, a_{jk}] = (-\hat{f}(k))^n a_{jk} \quad (207)$$

Therefore :

$$e^{iH'_0\lambda} a_{jk} e^{-iH'_0\lambda} = \sum_{n=0}^{+\infty} \frac{(-i\lambda\hat{f}(k))^n}{n!} a_{jk} = e^{-i\lambda\hat{f}(k)} a_{jk} \quad (208)$$

Finally we get

$$a_{jk} \frac{1}{\Delta E - H'_0} = \lim_{\epsilon \rightarrow 0} -i \int_0^{+\infty} d\lambda e^{i(\Delta E - H'_0 - \hat{f}(k) + i\epsilon)\lambda} a_{jk} \quad (209a)$$

$$a_{jk} \frac{1}{\Delta E - H'_0} = \frac{1}{\Delta E - H'_0 - \hat{f}(k)} a_{jk} \quad (209b)$$

A.4 Proof Of equation

Consider first the first term in equation 41, using equation 27 we get

$$\begin{aligned} \langle \Phi_f | a_{mp_1} V_{mp_1} \frac{1}{E_i - H_0 + i\epsilon} a_{np_2}^\dagger V_{np_2}^\dagger | \Phi_i \rangle &= \frac{1}{E_i - E_f - f(p_1, q) + i\epsilon} \langle \Phi_f | a_{mp_1} V_{mp_1} a_{np_2}^\dagger V_{np_2}^\dagger | \Phi_i \rangle \\ \langle \Phi_f | a_{mp_1} V_{mp_1} a_{np_2}^\dagger V_{np_2}^\dagger | \Phi_i \rangle &= \delta_{mn} \delta^{(3)}(p_1 - p_2) \langle \Phi_f | V_{mp_1} V_{np_2}^\dagger | \Phi_i \rangle + \langle \Phi_f | V_{mp_1} a_{np_2}^\dagger a_{mp_1} V_{np_2}^\dagger | \Phi_i \rangle \\ &= \delta_{mn} \delta^{(3)}(p_1 - p_2) \delta_{jl} \delta^{(3)}(k - q) \langle s', t' | V_{mp_1} V_{np_2}^\dagger | s, t \rangle + \delta_{ln} \delta^{(3)}(p_2 - q) \delta_m \delta^{(3)}(k - p_1) \langle s', t' | V_{mp_1} V_{np_2}^\dagger | s, t \rangle \end{aligned}$$

Caring out the integration we get

$$\begin{aligned} \sum_{n,m=1}^3 \int d^3p_1 \int d^3p_2 \langle \Phi_f | a_{mp_1} V_{mp_1} \frac{1}{E_i - H_0 + i\epsilon} a_{np_2}^\dagger V_{np_2}^\dagger | \Phi_i \rangle &= \\ \delta_{jl} \delta^{(3)}(k - q) \sum_{n=1}^3 \int d^3p_1 \frac{1}{E_i - E_f - f(p_1, q)} \langle s', t' | |V_{np_1}|^2 | s, t \rangle &+ \frac{1}{E_i - E_f - f(k, q)} \langle s', t' | V_{jk} V_{lq}^\dagger | s, t \rangle \\ = -(\delta_{jl} \delta_{ss'} \delta_{tt'} \delta^{(3)}(k - q) \frac{3f^2}{2(2\pi)^3 \mu_0^2} \int d^3p \frac{v(p^2)^2 p^2}{w_p f(p, q)} &+ \frac{1}{f(k, q)} \langle s', t' | V_{jk} V_{lq}^\dagger | s, t \rangle) \end{aligned}$$

The same calculation for the second term :

$$\sum_{n,m=1}^3 \int d^3p_1 \int d^3p_2 \langle \Phi_f | a_{mp_1}^\dagger V_{mp_1}^\dagger \frac{1}{E_i - H_0 + i\epsilon} a_{np_2} V_{np_2} | \Phi_i \rangle = \frac{1}{f(q, q)} \langle s', t' | V_{lq}^\dagger V_{jk} | s, t \rangle \quad (212)$$

A.5 Proof of equation 56

The expression of the probability density vector j is derived from the free Schrodinger equation :

$$i\hbar \frac{\partial \psi}{\partial t} = \frac{-\hbar^2}{2m} \nabla^2 \psi \quad (213)$$

Taking the complex conjugate we get

$$-i\hbar \frac{\partial \psi^*}{\partial t} = \frac{-\hbar^2}{2m} \nabla^2 \psi^* \quad (214)$$

Let the probability density $\rho(t) = \psi^* \psi$, taking the derivative of rho we get :

$$\frac{\partial \rho}{\partial t} = \frac{\partial \psi^*}{\partial t} \psi + \psi^* \frac{\partial \psi}{\partial t} \quad (215a)$$

$$\frac{\partial \rho}{\partial t} = \frac{-i\hbar}{2m} (\nabla^2 \psi^*) \psi + \frac{i\hbar}{2m} \psi^* (\nabla^2 \psi) \quad (215b)$$

$$\frac{\partial \rho}{\partial t} = \frac{\hbar}{2mi} ((\nabla^2 \psi^*) \psi - \psi^* (\nabla^2 \psi) + \nabla \psi \nabla \psi^* - \nabla \psi^* \nabla \psi) \quad (215c)$$

$$\frac{\partial \rho}{\partial t} = \frac{\hbar}{2mi} \nabla \cdot (\nabla \psi^* \psi - \psi^* \nabla \psi) \quad (215d)$$

Equation 215 defines a continuity equation for the probability density of the form

$$\frac{\partial \rho}{\partial t} + \nabla j = 0 \quad (216)$$

where

$$\vec{j} = \frac{-\hbar}{2mi} (\nabla \psi^* \psi - \psi^* \nabla \psi) \quad (217)$$

In spherical coordinates, the probability flux through a spherical surface is therefore (with the convention the flux inward in negative and outward is positive)

$$\Phi_R = - \oint \vec{j} \cdot \vec{ds} \quad (218)$$

with $\vec{ds} = R^2 d\Omega \vec{e}_r$ so that we get equation 56

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